To:	**M/s RAFCO GROUP**	**Ref:**	MES/PRO/2026-10-55
Attn:	Management / Projects Department	**Date:**	28 October 2026
Location:	Doha, State of Qatar	**Validity:**	10 calendar days

TECHNICAL & COMMERCIAL PROPOSAL

Design, Supply, Installation, Testing & Commissioning of a Cold Storage Facility – Five Freezer Rooms (−20 °C), Two Chiller Rooms (+2 °C) and Temperature-Controlled Corridor & Loading Lobby (+10 °C)

Dear Sir,

Thank you for the opportunity to submit our proposal for your new cold storage facility. Metri Engineering Services W.L.L. (MES), is pleased to offer a complete turnkey refrigeration package – from engineering design and insulated envelope to refrigeration plant, controls, testing and commissioning – based on the layout reviewed with your team.

We trust the proposal meets your requirements and remain at your disposal for any clarification or technical meeting.

1. Project Overview & Design Basis

Parameter	Basis
Facility	5 freezer rooms, 2 chiller rooms, 1 corridor with loading lobby (see Room Schedule, Section 2)
Design temperatures	Freezers −20/-18 °C · Chillers +2 °C · Corridor & loading lobby +10 °C
Outdoor design ambient	46 °C (Qatar summer design, condensing units rated accordingly)
Refrigeration concept	Each room served by 2 independent systems : DORIN ITALY condensing unit + LU-VE POLAND evaporator per system
Equipment selection basis	Freezers −27 °C evaporating · Chillers −5 °C evaporating · Ambient 46 °C
Walls & ceilings	150 mm PIR insulated sandwich panels, pre-painted steel both faces (Qatar made)
Floor insulation	XPS extruded polystyrene boards 75 mm, 2 layers as per temperature zone (Qatar made); concrete LAYER & epoxy by Client
Cold room doors	Manual sliding cold room doors 2.0 m W × 2.5 m H – 7 nos (one per freezer / chiller)

2. Room Schedule (BOQ Format)

Item	Room / Area	Design temp.	Internal size L × W (m)	Clear height (m)	Floor area (m²)	Volume (m³)	Sliding doors 2.0×2.5 m (nos)	Evaporators LU-VE (nos)	Cond. units DORIN (nos)	Remarks
1	Freezer Room F1	−20 °C	7.25 × 12.00	5.00	87.00	435.0	1	2	2	—
2	Freezer Room F2	−20 °C	7.25 × 12.00	5.00	87.00	435.0	1	2	2	—
3	Freezer Room F3	−20 °C	7.25 × 12.70	5.00	92.08	460.4	1	2	2	—
4	Freezer Room F4	−20 °C	7.25 × 12.70	5.00	92.08	460.4	1	2	2	—
5	Freezer Room F5	−20 °C	7.25 × 12.70	5.00	81.87	409.3	1	2	2	Less lift shaft 3.19 × 3.20 m
6	Chiller Room C1	+2 °C	7.25 × 12.00	8.00	87.00	696.0	1	2	2	—
7	Chiller Room C2	+2 °C	4.95 × 12.00	8.00	59.40	475.2	1	2	2	—
8	Corridor	+10 °C	27.15 × 4.50	5.00	122.18	610.9	TO BE DECIDED LATER BY CLIENT	2	2	TO BE DECIDED LATER BY CLIENT
9	Loading Lobby (dock D6)	+10 °C	4.95 × 5.40	5.00	26.73	133.7	TO BE DECIDED LATER BY CLIENT	—	—	TO BE DECIDED LATER BY CLIENT
	TOTAL						**7**	**16**	**16**	

3. Bill of Quantities – Supply & Installation

Item	Description	Unit	Qty	Remarks
A	**ENGINEERING**			
A.1	Complete engineering design: heat load calculations, equipment selection, refrigerant piping design, control philosophy and shop drawings	Lot	1	For Client approval before procurement
B	**INSULATED ENVELOPE**			
B.1	PIR insulated sandwich panels 150 mm, walls – Qatar made, 0.5mm thick PPGI, incl. base channels, corner profiles, flashings & sealants	m²	1,351	Net installed area (approx.)

Item	Description	Unit	Qty	Remarks
B.2	PIR insulated sandwich panels 150 mm, ceilings – Qatar made, incl. ceiling-to-wall profiles, suspension hangers & rods (to Client steel structure)	m²	770	Net installed area (approx.)
B.3	XPS extruded polystyrene floor insulation boards 75 mm, 2 layers as per temperature zone – Qatar made	m²	735	Concrete topping & epoxy by Client
B.4	Manual sliding cold room doors 2.0 m W × 2.5 m H complete with frame, top track, hardware and inside safety release	Nos	7	One per freezer / chiller room
C	**REFRIGERATION SYSTEMS**			
C.1	DORIN (Italy) air-cooled condensing units – low temperature, freezer rooms F1–F5	Nos	10	2 per room
C.2	DORIN (Italy) air-cooled condensing units – medium temperature, chiller rooms C1–C2	Nos	4	2 per room
C.3	DORIN (Italy) air-cooled condensing units – corridor & loading lobby	Nos	2	
C.4	LU-VE (Europe) evaporators (unit coolers) matched to condensing units, with electric defrost for freezer units	Nos	16	2 per room / corridor
C.5	Expansion valves, solenoid valves, filter driers, sight glasses and refrigeration accessories	Lot	1	
C.6	Refrigerant copper piping with insulation, supports, refrigerant charge, oil and drain line heaters for freezer evaporators	Lot	1	
D	**ELECTRICAL, CONTROLS & SAFETY**			
D.1	Refrigeration control panels with electronic room controllers, temperature display and alarms	Lot	1	main isolator & cable (by others)
D.2	LED cold-rated lighting for cold rooms and corridor	Lot	1	
E	**TESTING, COMMISSIONING & HANDOVER**			
E.1	Pressure testing, evacuation, charging, testing & commissioning of all systems	Lot	1	
E.2	As-built drawings, O&M manuals and operator training	Lot	1	
	TOTAL CONTRACT VALUE (LUMP SUM)			**QAR 1,750,250**

Quantities of panels and floor insulation are approximate, derived from the layout drawing, and form part of the lump sum; final quantities will follow the approved shop drawings and site measurements.

4. Scope of Work

1. Complete engineering design including heat load calculations, equipment selection, refrigerant piping, and control requirements.
2. Supply and installation of PIR insulated sandwich panels 15 cm thickness for all walls and ceilings, made in Qatar.
3. Supply and installation of XPS floor insulation polystyrene boards made in Qatar (75 mm, 2 layers as per temperature zone). Concrete on top and epoxy by Client.
4. Supply and installation of manual sliding cold room doors (2.0 m W × 2.5 m H) for all cold rooms – total 7 nos doors.
5. Supply, installation and commissioning of condensing units made by DORIN ITALY, with matching evaporators made by LU-VE Europe, with all necessary expansion valves, controls and refrigerant piping.
6. Complete lighting (LED cold-rated), control panels and safety systems.
7. Testing & commissioning of all systems.
8. As-built drawings, O&M manuals and operator training.
9. Warranty: 400 days.

5. Contract Value

Total Contract Value (Lump Sum)	**QAR 1,750,250**
In words	*Qatari Riyals One Million Seven Hundred Fifty Thousand Two Hundred Fifty Only*

Main exclusions

- Civil works and building structure and any steel supports
- Epoxy flooring
- Concrete works (slab, foundations, screed)
- Painting
- Building permits and approvals (by Client)
- Electrical supply up to the main isolator near our indoor / outdoor units
- Fire suppression systems
- Racking / shelving inside cold rooms

6. Commercial Terms & Conditions

6.1 Payment Terms

To be discussed and agreed.

6.2 Validity

This proposal is valid for 10 calendar days from the date of submission.

6.3 Delivery & Installation Period

Estimated project duration is 14-16 weeks from contract signing and receipt of advance payment, subject to site readiness (completed civil works, slab and building envelope).

7. Warranty

MES shall provide a comprehensive warranty of 400 days from successful commissioning covering all supplied equipment, materials and workmanship. Post-warranty Annual Maintenance Contract (AMC) offered at competitive rates.

8. Exclusions & Clarifications

Our scope does not include civil works, removal of existing units, steel base frames / foundation for condensers, machine room, closed working room and storage on site, electrical power supply on site during installation, plumbing and sanitary works, electrical sub-distribution panel in machinery room, electrical connection between sub-distribution panel and our control panels, insurance policy, governmental fees, any other related additional costs and any supplies and services not especially listed herein.

Without limitation, the following items are explicitly excluded:

1. All kinds of civil works including passage / sleeve for refrigerant pipes through the walls, etc.
2. Electrical power supply and earthing to our control panels.
3. Drain points as required.
4. Foundation or steel platforms for the condensers.
5. Adequate safe storage area for all equipment and materials.
6. Temporary water and electricity during installation.
7. Disposal and carting away of sundry items.
8. Offloading of materials at site.
9. Any third-party approvals.
10. Factory visits.
11. Dosing system (water treatment) for the evaporative condensers.
12. Any other item / service not specifically mentioned herein.
13. Any special specifications other than the standard.
14. Any pipe ventilation works.
15. Concrete works or floor finishes (concrete topping over XPS is by others).
16. Any steel platform to hang the outdoor / indoor units.
17. Any civil works and plinths.
18. Any electrical works (power supply, cabling, distribution boards, earthing – all by Electrical Contractor).
19. Fire detection and firefighting systems within cold rooms.
20. Building Management System (BMS) integration.
21. Racking and shelving systems within cold rooms.
22. Floor heating system for frost-heave prevention.

9. Layout Drawing

Refrigeration equipment and sliding door layout – plan (MES sheet E-01).

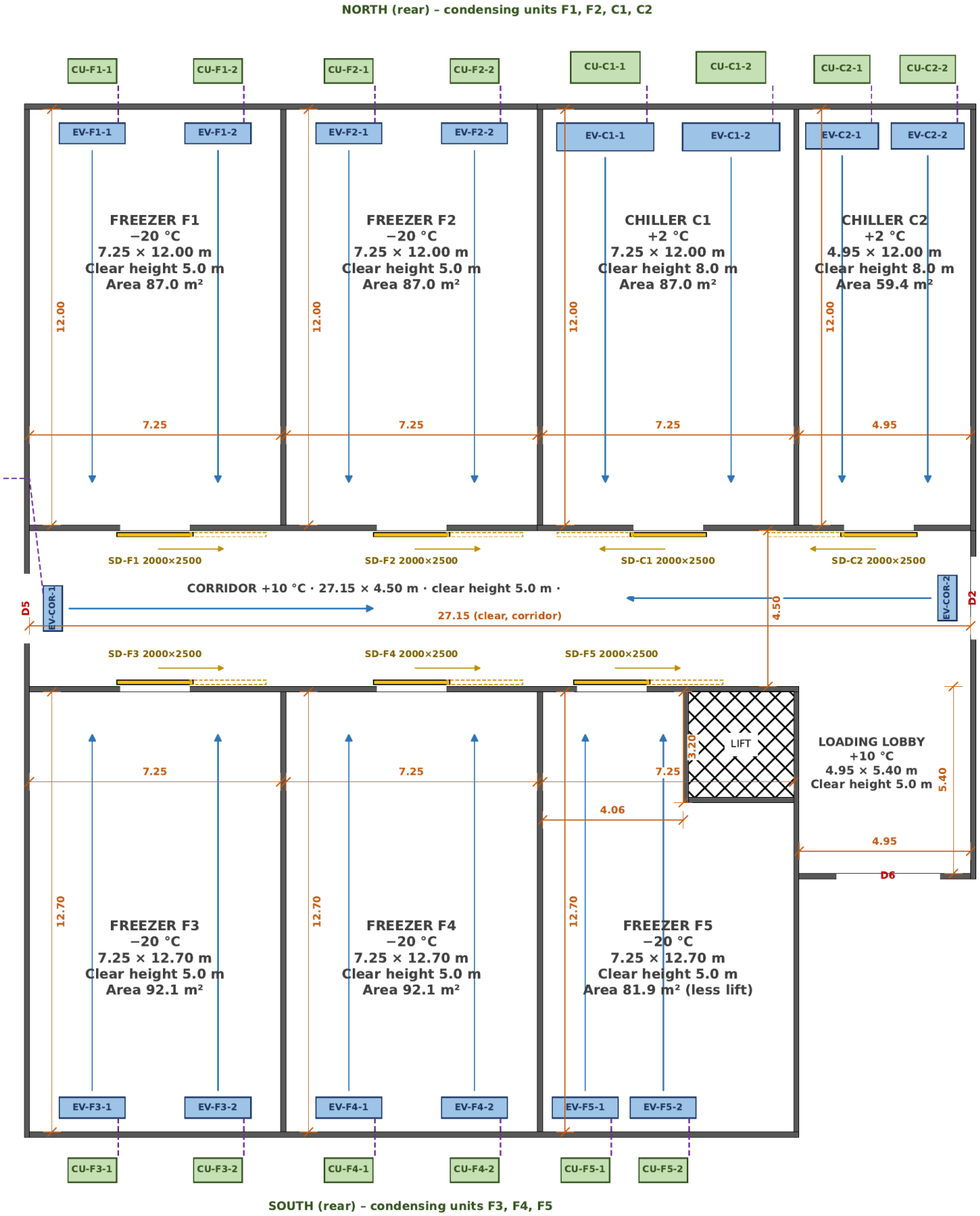


Figure 1 – Room layout with dimensions, clear heights, sliding doors, evaporators (EV) and condensing units (CU)

10. 3D Visualisation

Concept renders of the proposed facility. Equipment is shown indicatively; final appearance per approved shop drawings and selected models.

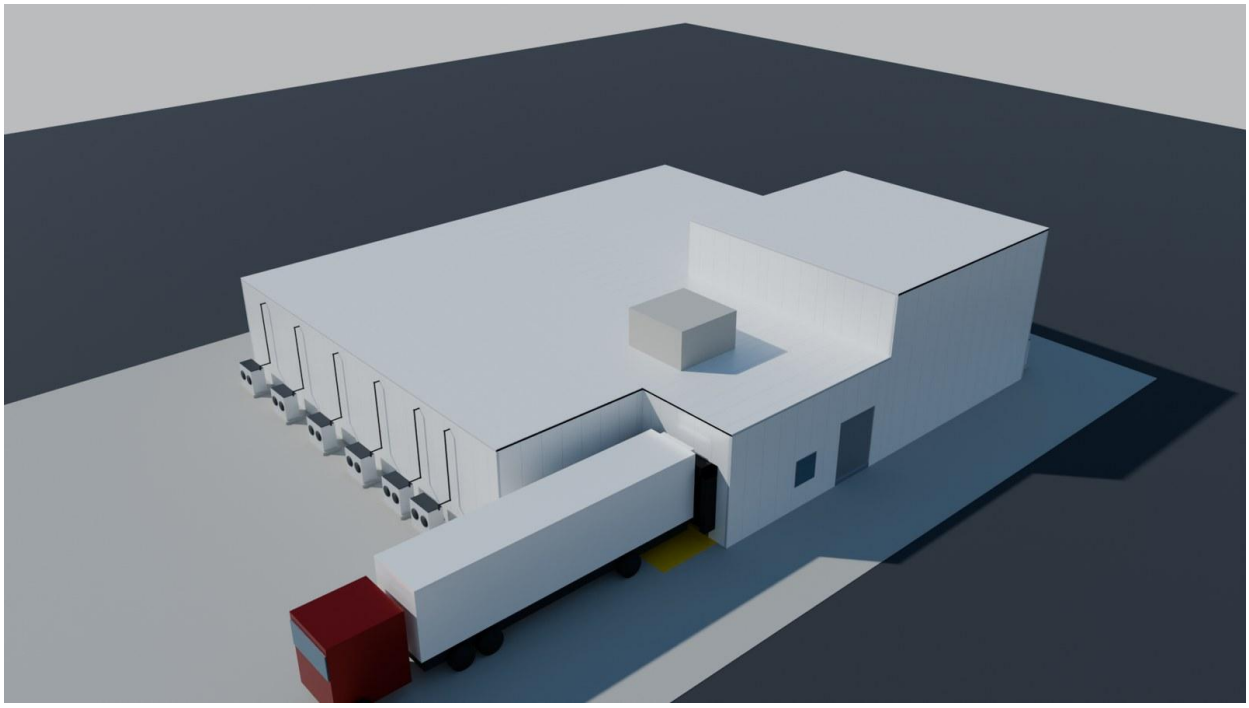


Figure 2 – Aerial view from south-east: loading dock, freezer condensing units on the south wall, 8 m chiller block behind

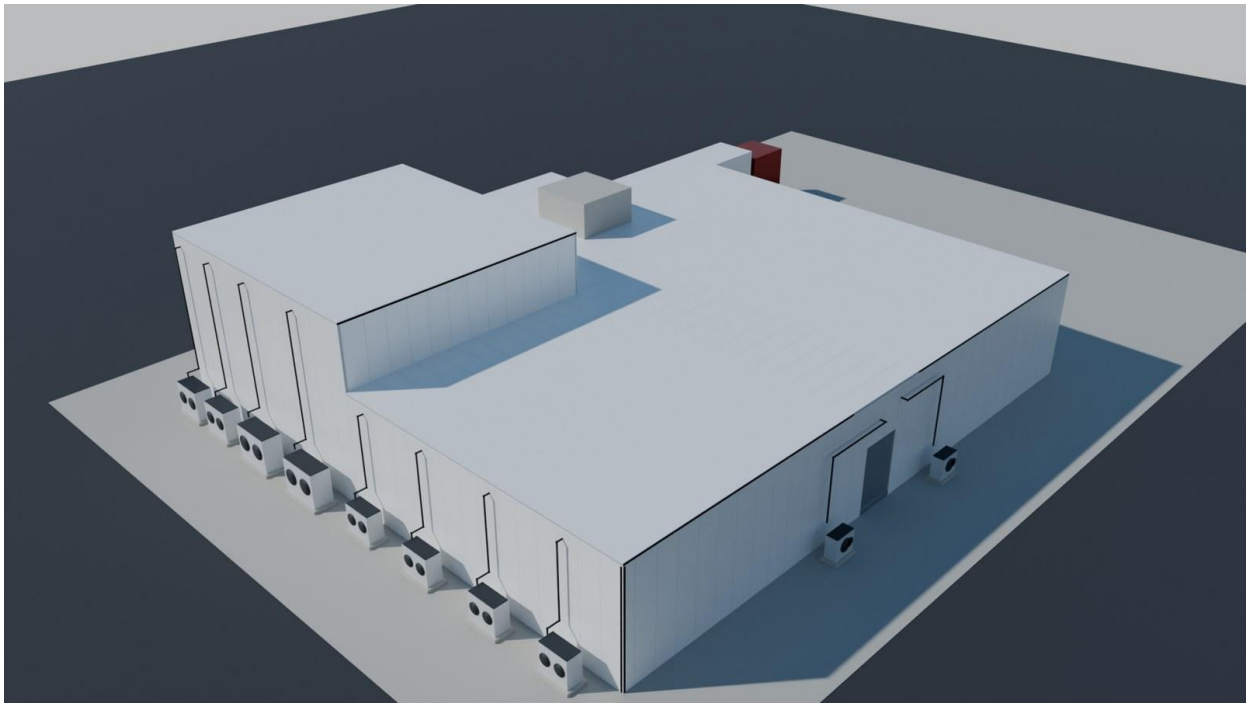


Figure 3 – Aerial view from north-west: chiller and freezer condensing units on the north rear wall

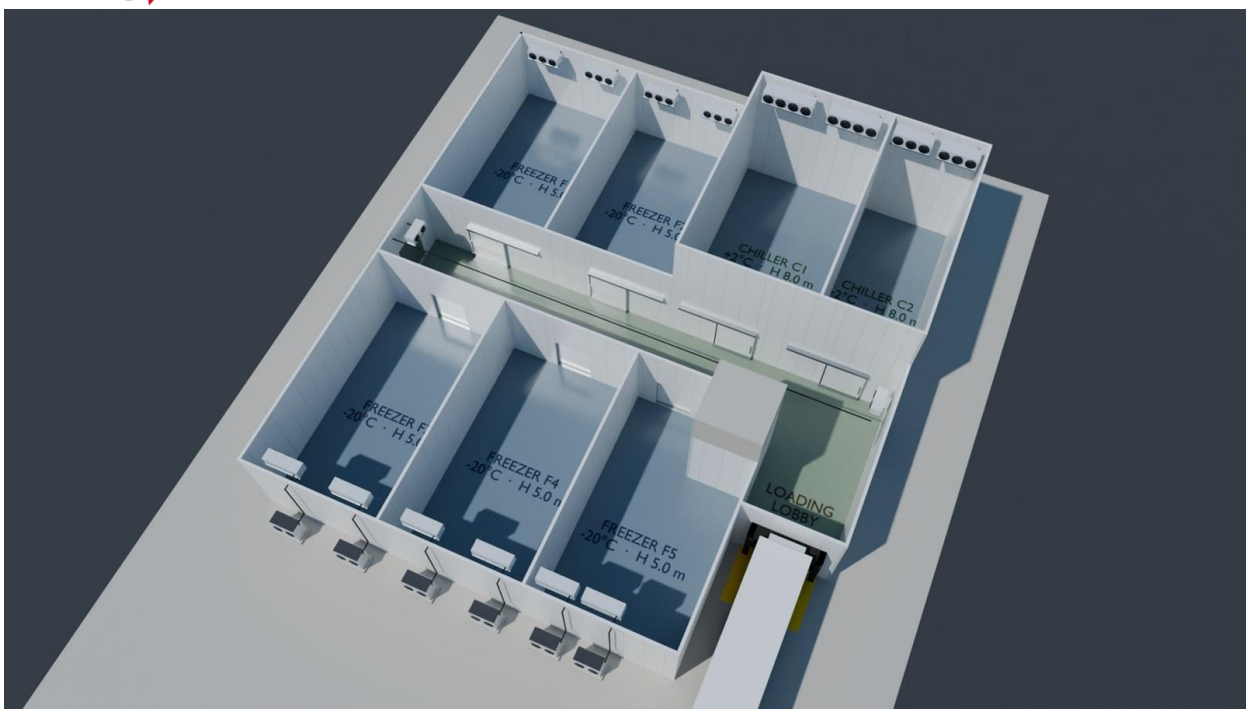


Figure 4 – Cut-away view (ceilings removed): two evaporators per room on the rear wall, sliding doors to the corridor



Figure 5 – Corridor interior (+10 °C): manual sliding cold room doors 2.0 × 2.5 m

We trust our offer is in line with your requirements. For any clarification please do not hesitate to contact us.

METRI ENGINEERING SERVICES W.L.L.
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